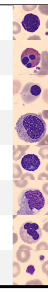
Table 1: Class-wise statistics and representative samples for Blood. The layout expands the corresponding row in the main-paper dataset table: the class-count field is replaced by individual class names. Samples are taken directly from the training split.

Dataset	Class	Train	Val	Test	Total	Sample
	Basophil	852	122	244	1,218	
	Eosinophil	2,181	312	624	3,117	
	Erythroblast	1,085	155	311	1,551	
	Immature granulocytes	2,026	290	579	2,895	
Blood	Lymphocyte	849	122	243	1,214	
	Monocyte	993	143	284	1,420	
	Neutrophil	2,330	333	666	3,329	
	Platelet	1,643	235	470	2,348	

Supplementary Results and Analysis

Detailed Dataset Description

The experiments use five MedMNIST v2 benchmarks (Yang, Shi, and Ni 2023). They were selected to cover microscopy, dermoscopy, CT, and ultrasound while varying the number of diagnostic classes and the prevalence profile. Table ?? reports the exact split sizes used by the experiments; the descriptions below clarify the recognition problem represented by each benchmark.

Blood. BloodMNIST is an eight-class microscopy benchmark for recognizing peripheral blood-cell morphology. Its classes include granulocytes, lymphocytes, and other cells with visually similar staining patterns. Fine-grained morphology makes it a useful stress test for whether a merged classifier preserves class-specific directions rather than concentrating its predictions on a small subset of cells.

Derma. DermaMNIST contains seven dermoscopic skin-lesion categories. Its prevalence is strongly imbalanced and several categories share colour and texture patterns. This setting tests whether a merger distinguishes rare lesions without erasing the clinically meaningful prevalence of common lesions.

Organ-C and Organ-S. OrganCMNIST and OrganSMNIST provide 11-way abdominal CT organ recognition in coronal and sagittal views, respectively. The two views share the same broad task but change the visible anatomy and image geometry, testing robustness across CT planes.

Ultrasound. UltrasoundMNIST is an eight-class ultrasound-structure benchmark. Speckle noise, low contrast, and view-dependent boundaries make it particularly suitable for the prediction-collapse diagnostics and the per-class gap visualization in Fig. ??.

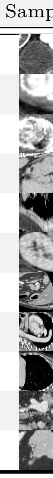
Table 2: Class-wise statistics and representative samples for Derma, using the same format as Table 1.

Dataset	Class	Train	Val	Test	Total	Sample
	Actinic keratosis / carcinoma	228	33	66	327	
	Basal cell carcinoma	359	52	103	514	
	Benign keratosis-like lesion	769	110	220	1,099	
Derma	Dermatofibroma	80	12	23	115	
	Melanoma	779	111	223	1,113	
	Melanocytic nevus	4,693	671	1,341	6,705	
	Vascular lesion	99	14	29	142	

Table 3: Class-wise statistics and representative samples for Organ-C.

Dataset	Class	Train	Val	Test	Total	Sample
	Bladder	1,148	191	828	2,167	
	Femur–left	619	102	431	1,152	
	Femur–right	595	96	421	1,112	
	Heart	600	202	421	1,223	
	Kidney–left	1,088	132	727	1,947	
	Kidney–right	1,170	157	735	2,062	
Organ-C	Liver	2,986	429	1,835	5,250	
	Lung–left	1,002	347	549	1,898	
	Lung–right	1,022	352	557	1,931	
	Pancreas	1,173	179	750	2,102	
	Spleen	1,572	205	962	2,739	

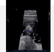
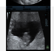
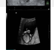
Table 4: Class-wise statistics and representative samples for Organ-S.

Dataset	Class	Train	Val	Test	Total	Sample
	Bladder	1,148	188	811	2,147	
	Femur–left	630	104	439	1,173	
	Femur–right	614	95	445	1,154	
	Heart	721	246	510	1,477	
	Kidney–left	1,132	140	704	1,976	
	Kidney–right	1,119	159	693	1,971	
Organ-S	Liver	3,464	491	2,078	6,033	
	Lung–left	741	261	397	1,399	
	Lung–right	803	275	439	1,517	
	Pancreas	2,004	280	1,343	3,627	
	Spleen	1,556	213	968	2,737	

Detailed Ablation Configurations and Interpretation

All internal ablations use the same five datasets, four backbones, client counts, and Dirichlet settings as the

Table 5: Class-wise statistics and representative samples for Ultrasound. The source benchmark retains anonymized class identifiers, which are reported without assigning unverified clinical names.

Dataset	Class	Train	Val	Test	Total	Sample
Ultrasound	Class 0	420	60	121	601	
	Class 1	419	59	121	599	
	Class 2	408	58	117	583	
	Class 3	630	90	180	900	
	Class 4	345	49	100	494	
	Class 5	489	69	141	699	
	Class 6	671	95	193	959	
	Class 7	487	69	140	696	

main experiment. Thus, a change in Table ?? or Table ?? isolates one design choice while holding the remaining LAMP-Merge pipeline fixed.

Diagnostic Prototype Reconstruction For class c, DPR aggregates reference-space prototypes as

$$p_c = \sum_{i=1}^{K} \alpha_{i,c}\mu_{i,c}, \qquad \alpha_{i,c} = \frac{e_{i,c}}{\sum_j e_{j,c}}. \tag{1}$$

$$e_{i,c} = (n_{i,c}+1)^{\gamma}\mathbf{1}[n_{i,c} > 0]. \tag{2}$$

Here $\mu_{i,c}$ is client i's class-conditional reference-space mean and $n_{i,c}$ is its support for that class. The substitutions below test whether the class semantics in $\mu_{i,c}$ and the class-specific reliability in $\alpha_{i,c}$ are both required.

Classifier-head aggregation (CHA). CHA replaces the class prototype in Eq. 1, $\mu_{i,c} \longrightarrow h_{i,c}$, with the local classifier-head direction. It tests whether client heads already provide a common, class-aligned representation. Its mean ACC is 25.63%, a 36.58-point reduction from LAMP-Merge (62.21%), showing that independently trained head directions are not a reliable substitute for shared-space class evidence.

Global-feature mean (GFM). GFM replaces the class-conditional prototype by the same class-agnostic feature mean for every class, $\mu_{i,c} \longrightarrow g$. This tests whether generic client appearance alone can drive merging. Its 26.50% mean ACC is 35.71 points below LAMP-Merge, indicating that class conditioning, not merely feature aggregation, is essential.

Support-only synthetic head (SSH). SSH replaces the aggregated prototype $p_c = \sum_i \alpha_{i,c}\mu_{i,c}$ by $p_c \longrightarrow u_c$, where u_c is a fixed random unit vector. It preserves the classifier interface while eliminating diagnostic content. The 10.89% mean ACC, 51.32 points below LAMP-Merge, verifies that performance does not arise from support counts or the cosine-head form alone.

Shuffled-label prototype (SLP). SLP replaces the class-aligned aggregate by $p_c \longrightarrow p_{\sigma(c)}$, for a permutation σ of the diagnosis labels. Its 19.57% mean ACC (a 42.64-point reduction) directly tests semantic alignment: useful class evidence must be assigned to the correct output class.

Uniform client weight (UCW). UCW replaces the support-aware weight $e_{i,c} = (n_{i,c}+1)^{\gamma}\mathbf{1}[n_{i,c} > 0]$ by equal class-presence evidence:

$$e_{i,c} \longrightarrow \mathbf{1}[n_{i,c} > 0], \qquad \alpha_{i,c} = \frac{e_{i,c}}{\sum_j e_{j,c}}. \tag{3}$$

It tests whether observing a class is sufficient, independent of the amount of evidence. Its 58.33% mean ACC is 3.88 points below LAMP-Merge, showing that support magnitude provides useful reliability information beyond class presence.

Binary support only (BSO). BSO replaces both class support and prevalence counts by presence indicators, $n_{i,c}, m_{i,c} \longrightarrow \mathbf{1}[n_{i,c} > 0]$. It tests whether fine-grained class frequency is necessary in both modules. The mean ACC falls to 55.18% (7.03 points below LAMP-Merge), the largest loss among the weighting controls.

Global client-size weight (GCSW). GCSW replaces class-specific support in the reliability weight by total client size:

$$e_{i,c} = (n_{i,c}+1)^{\gamma}\mathbf{1}[n_{i,c} > 0] \longrightarrow (N_i+1)^{\gamma}\mathbf{1}[n_{i,c} > 0]. \tag{4}$$

Here $N_i = \sum_k n_{i,k}$ is the total support of client i. GCSW tests whether a generally large client can stand in for strong evidence of a particular class. Its 59.39% mean ACC, 2.82 points below LAMP-Merge, shows that local data volume should be assigned at the class level.

DPR summary. The semantic controls cause large reductions (35.71–51.32 points), while the weighting controls cause smaller but consistent reductions (2.82–7.03 points). Together, these results support the two parts of Eq. 1: class-conditioned shared-space evidence provides the discriminative direction, and class-specific support determines how reliably each client contributes to that direction.

Long-tail Prevalence Calibration LPC estimates an empirical prior and adds a centered, gated log-prior bias after DPR:

$$\pi_c = \frac{\sum_i m_{i,c}}{\sum_k \sum_i m_{i,k}}, \qquad b_c = \lambda \left(\log \pi_c - \frac{1}{C} \sum_k \log \pi_k \right). \tag{5}$$

$$\ell_c(x) = w_c^{\top} \phi_0(T(x)) + \mathbf{1}[r > \tau] b_c. \tag{6}$$

The three controls retain DPR and test the prior estimator or the activation gate in Eq. 5.

Uniform prevalence prior (UPP). UPP replaces the empirical prior in Eq. 5 by $\pi_c \longrightarrow 1/C$. It tests whether calibration should ignore the observed class distribution. Mean ACC is 58.91%, 3.30 points below LAMP-Merge, demonstrating that the empirical prior is useful rather than a source of collapse.

Always-on calibration (AOC). AOC replaces the activation gate in the score $\ell_c(x)$ by $\mathbf{1}[r > \tau] \longrightarrow 1$. It tests whether the prior should be applied uniformly to every configuration. Its 61.90% mean ACC is 0.31 points below LAMP-Merge. The small aggregate gap indicates that calibration is broadly useful, while the gated design avoids imposing a prior when a configuration is not sufficiently long-tailed.

Client-balanced prior (CBP). CBP replaces the sample-count empirical prior by equal client influence:

$$\pi_c = \frac{\sum_i m_{i,c}}{\sum_k \sum_i m_{i,k}} \longrightarrow \frac{1}{K} \sum_i \frac{m_{i,c}}{\sum_k m_{i,k}}. \tag{7}$$

It tests whether each client should contribute equally regardless of sample volume. The 58.97% mean ACC, 3.24 points below LAMP-Merge, favors the sample-count empirical prior over a client-balanced estimate.

LPC summary. Every LPC control is below the 62.21% LAMP-Merge mean. UPP and CBP show that both prevalence information and its sample-size weighting matter; AOC shows that the activation criterion is a targeted safeguard rather than the primary source of the gain. Hence LPC calibrates DPR's restored class directions instead of replacing them with a prior.

Additional Analysis of DPR and LPC

Prototype estimation and class margins Let $\mu_c^\star$ be the population class mean in the shared reference space. For client-level class estimates with bounded second moments, the mean-squared error of the aggregated prototype obeys

$$\mathrm{E}\|p_c - \mu_c^\star\|_2^2 \le \sum_i \alpha_{i,c}^2 \frac{\sigma_c^2}{n_{i,c}} + \mathrm{Bias}_c^2. \tag{8}$$

The first term is a variance contribution: it decreases with support $n_{i,c}$, but only clients that observed class c may contribute. The second term represents systematic mismatch between a local class estimate and the shared-space population mean. Equation 8 explains both support-related controls: UCW ignores unequal estimation variance, whereas GCSW confuses total client size with the support relevant to class c.

For a reference feature $z = \phi_0(T(x))$, consider two candidate classes c and d. The population margin is

$$\Delta_{c,d}(x) = (\mu_c^\star)^\top z - (\mu_d^\star)^\top z. \tag{9}$$

If

$$\|p_c - \mu_c^\star\|_2 + \|p_d - \mu_d^\star\|_2 < \frac{\Delta_{c,d}(x)}{\|z\|_2}, \tag{10}$$

then the reconstructed prototypes preserve the ordering between c and d. Thus, class-specific aggregation is not intended to reproduce arbitrary local heads: it reduces prototype error sufficiently to retain discriminative class margins. The large losses for CHA, GFM, SSH, and SLP are consistent with breaking precisely this semantic margin mechanism.

Bounded prevalence correction The centering in Eq. 5 removes the uniform component of the log prior. Therefore only relative scores change, with

$$b_c - b_d = \lambda \log \frac{\pi_c}{\pi_d}. \tag{11}$$

The gate $\mathbf{1}[r > \tau]$ applies this relative correction only under sufficient prevalence imbalance, and the finite λ bounds its magnitude. DPR consequently supplies the class directions, while LPC adjusts the decision boundary only when the empirical class prevalence provides relevant evidence. This division also explains why a prevalence-only mechanism cannot solve the semantic controls in Table ??.

Metric Definitions

Let y_n and $\hat{y}_n$ denote the true and predicted labels of sample n, respectively. Let $q_c = N^{-1} \sum_n \mathbf{1}[\hat{y}_n = c]$ be the predicted class distribution and $p_c = N^{-1} \sum_n \mathbf{1}[y_n = c]$ the true distribution. The prediction-collapse diagnostics are

$$\rho = \max_c q_c, \qquad C_{\text{eff}} = \exp\left(-\sum_c q_c \log q_c\right). \tag{12}$$

$$\mathrm{TV}(q, p) = \frac{1}{2} \sum_c |q_c - p_c|. \tag{13}$$

The collapse ratio ρ measures the largest predicted-class share, C_{eff} measures the effective number of classes used by the predictor, and TV measures distributional disagreement with the labels. Lower ρ and TV and higher C_{eff} are preferred. The absolute per-class gaps in Fig. ?? are $|q_c - p_c|$ in percentage points and expose which classes contribute to TV.

Hyperparameter-Sweep Reproducibility

Every point in Fig. ?? is a completed full-scope experiment covering five datasets, four backbones, three client counts, and three Dirichlet settings. The plotted curves retain measured points in ascending parameter order; no smoothing, interpolation, or best-run selection is applied. The LPC grid records $\tau \in \{2.0, 2.5, 3.0\}$ and calibration strengths $\lambda \in \{3.00, 3.25, \ldots, 5.25\}$, with $\gamma = 0.55$ and $s = 18.75$ fixed. DPR completion records are plotted only after the full grid point has succeeded. The selected operating point is $\gamma = 0.55$, $s = 18.75$, $\tau = 2.5$, and $\lambda = 4.25$, making its reported value traceable to the same completed grid as its neighboring points.

References

Yang, J.; Shi, R.; and Ni, B. 2023. MedMNIST v2: A Large-Scale Lightweight Benchmark for 2D and 3D Biomedical Image Classification. Scientific Data, 10(1): 41.